

Phonetic Prior Encoder

Training with
ASR Transcript

Inference with
Lips Input

Phonemes

P_1

P_2

P_3

OR

Special Token

$\langle global \rangle$

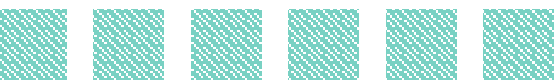
Text Encoder

KV

Cross Attention

FFN

Q



Phonetic Prior Embeddings (Λ)

Phonetic
Priors



Prior-Fused Visual Encoder (PFVE)

Lips
Input



Lip Encoder



Priors



f_V^0

KV

Cross Attention

Q

Conformer Layer

f_V^{l-1}

R^l

$\hat{f}_V^l$

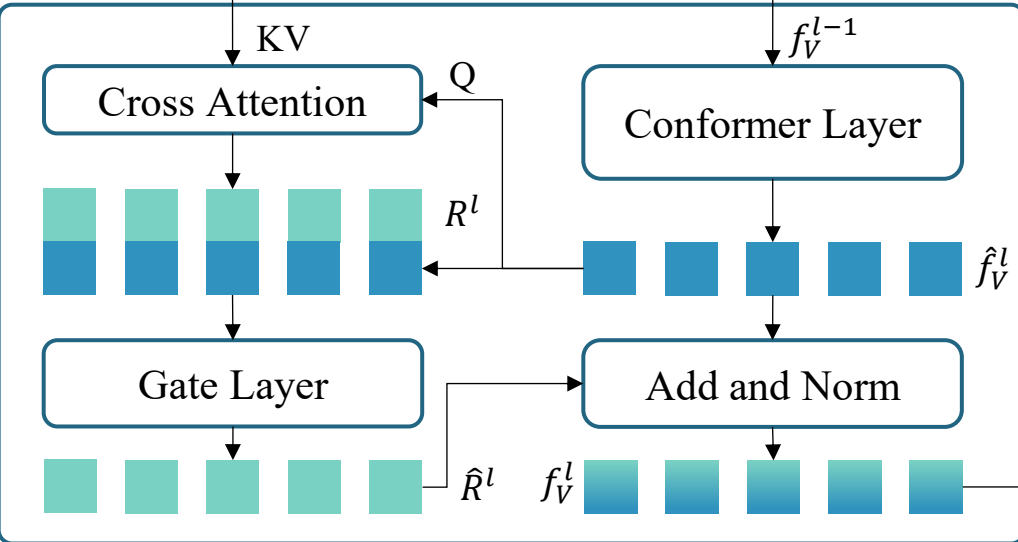
Gate Layer

Add and Norm

$\hat{R}^l$

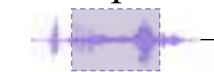
f_V^l

$L_C \times$



Speech Generator (SG)

Ref Speech



Speaker
Encoder

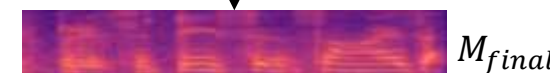


h_{spk}

Acoustic Convertor



CFM Module



Vocoder



Synthesized Speech